Article

Introducing an evolutionary algorithm for the optimal training of RBF networks

Ioannis G. Tsoulos[1,*], Vasileios Charilogis[2] and Dimitrios Tsalikakis[3]

[1] Department of Informatics and Telecommunications, University of Ioannina, 451 10 Ioannina, Greece; email: itsoulos@uoi.gr
[2] Department of Informatics and Telecommunications, University of Ioannina, 451 10 Ioannina, Greece; email: v.charilog@uoi.gr
[3] Department of Engineering Informatics and Telecommunications, University of Western Macedonia, 50100 Kozani, Greece; email: dtsalikakis@uowm.gr
* Correspondence: itsoulos@uoi.gr

Abstract

A large collection of real - world classification and regression problems can be addressed using machine learning tools such as, for example, radial basis function networks (RBF networks). However, the techniques used for training RBF networks often exhibit various problems, such as getting trapped in local minima of the error function, or even encountering numerical issues when solving systems of linear equations in order to estimate the parameters of the RBF network. This paper presents a multi-stage evolutionary technique based on genetic algorithms for the effective training of RBF networks. In the first stage, the value ranges of the RBF network parameters are estimated using the K-Means algorithm. In the second stage, the chromosomes of the genetic algorithm are initialized within the parameter ranges determined in the first stage, followed by the execution of the genetic algorithm. Each chromosome of the genetic algorithm is considered a candidate parameter vector for the machine learning model. The centers and variances of the RBF network are estimated by the Genetic Algorithm, while the network weights are determined by solving a system of linear equations. This method was applied to a large set of classification and data-fitting problems, yielding excellent results.

Keywords: Machine learning; neural networks; genetic algorithms.

1. Introduction

The rapid development of machine learning has enabled its adoption in numerous practical applications. Representative examples can be found in scientific and technological fields such as physics [1,2], astronomy [3,4], chemistry [5,6], medicine [7,8], economics[9,10], image processing [11], time-series prediction [12], financial and credit card analysis [13], and mechanical engineering [14]. Such problems are typically tackled through machine learning approaches, with Radial Basis Function (RBF) networks [15,16] representing one of the most widely adopted and effective model classes. An RBF network is commonly defined as a function:

$$R(\overrightarrow{x}) = \sum_{i=1}^{k} w_i \phi(\| \overrightarrow{x} - \overrightarrow{c_i} \|) \tag{1}$$

Received:
Revised:
Accepted:
Published:

https://doi.org/10.3390/1010000

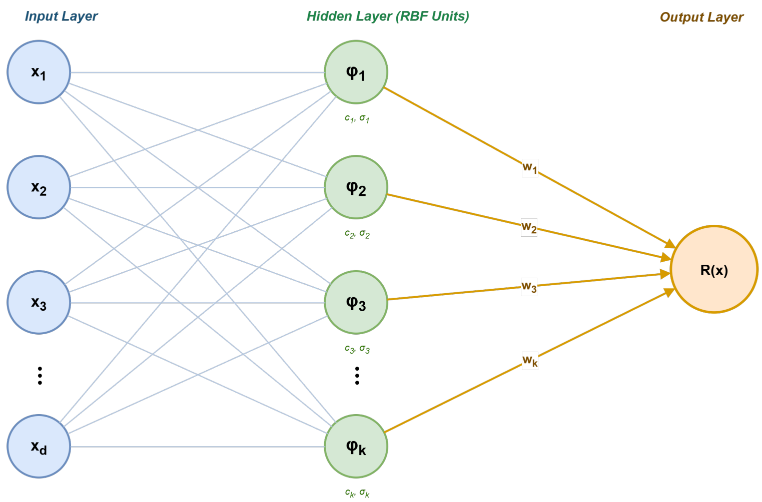

Figure 1. General structure of an RBF network. The input vector $x = (x_1, x_2, ..., x_d)$ is fed to k hidden units, each characterized by a center c_i (a reference point in the input space) and a variance σ_i (the spread of the basis function). The output $R(x)$ is a weighted sum of the hidden unit responses, with weights w_i.

The following notation is used in Equation 1:

1. The vector $\overrightarrow{x}$ denotes the input pattern and the number of elements in this vector is denoted as d.
2. The constant k denotes the number of weights for the RBF network.
3. The vectors $\overrightarrow{c_i}$, $i = 1, .., k$ stand for the so - called center vectors.
4. The vector $\overrightarrow{w}$ represents the weight vector of the RBF network.
5. The value $R(\overrightarrow{x})$ denotes the predicted value of the network for the input pattern $\overrightarrow{x}$.
6. The function $\phi(x)$ typically is defined as the following Gaussian function:

$$\phi(x) = \exp\left(-\frac{(x-c)^2}{\sigma^2}\right) \tag{2}$$

Moreover, the training error associated with any RBF network $R(x)$ is defined as:

$$E(R(\overrightarrow{x})) = \sum_{i=1}^{M} \left(R(\overrightarrow{x_i}) - y_i\right)^2 \tag{3}$$

In this equation, the set $(\overrightarrow{x_i}, y_i)$, $i = 1, ..., M$ represents the training set of the objective problem and the values y_i are the actual outputs for any given pattern $\overrightarrow{x_i}$. The effectiveness of RBF networks has been demonstrated across numerous application domains. Representative examples include physics-related problems [17–20], differential equation solving [21–23], robotic systems [24,25], facial recognition [26], digital communication systems [27,28], chemical applications [29,30], economic modeling [31–33], cybersecurity and network security tasks [34,35], time-series forecasting [36], and wind energy production prediction [37].

In recent years, numerous studies have introduced novel approaches for initializing the parameters of RBF networks [38–40]. In addition, the influence of kernel widths on the performance of RBF networks has been extensively investigated by Benoudjit et al. [41], while Neruda et al. [42] provided a comparative evaluation of several learning algorithms designed for RBF training. Furthermore, a variety of pruning strategies have been proposed [43–45] with the objective of reducing the number of network parameters and, consequently, the overall model complexity. Given the widespread adoption of RBF networks and the substantial computational effort often required for their training, several studies have recently focused on exploiting parallel computing architectures to accelerate neural network parameter optimization [46,47]. Finally, Park et al. [48] demonstrated that an RBF network consisting of a single hidden processing layer possesses the universal approximation property, being capable of approximating any continuous function with arbitrary accuracy.

In conventional RBF network training, the centers of the basis functions $\phi(x)$ are initially determined, and the corresponding weight vector $\overrightarrow{w}$ is subsequently obtained by solving a linear system of equations. The k-means clustering algorithm [49] is among the most frequently employed techniques for center selection. Despite its simplicity and computational efficiency, this approach often suffers from overfitting, reducing the predictive performance of the model on unseen samples. Moreover, because the optimization process does not impose explicit constraints on the parameter values, the resulting estimates may become excessively large or small, adversely affecting both model stability and generalization capability.

The present work introduces a novel training methodology for RBF networks consisting of two distinct phases. During the first phase, a promising range of values for the network centers and variances is estimated by exploiting the k-means clustering technique. In the second phase, a modified genetic algorithm [50,51] is employed to determine the optimal values of the centers and variances within the search intervals identified during the first phase. The weights of the RBF network are computed through the solution of an appropriate linear system of equations.

The remaining of this article is organized as follows: in section 2 the proposed method is fully described, in section 3 the experimental results are outlined and finally in section 4 a discussion is provided on the experimental results.

2. Materials and Methods

2.1. The first phase of the proposed method

In the first phase, the k-means clustering algorithm is employed to estimate promising search intervals for the centers and variances of the RBF network. Specifically, for each network parameter, the corresponding search interval is defined as a multiple of the value produced by the k-means algorithm, thereby providing a bounded region for the subsequent optimization process. The k-means algorithm is presented in Algorithm 1.

https://doi.org/10.3390/1010000

Algorithm 1 The k-means algorithm used to estimate centers and variances.

1. **Initialization**
   (a) **Input**: the train set $\left(\overrightarrow{x_i}, y_i\right),\ i = 1, ..., M$ of the objective problem.
   (b) **Set** as k the number of desired centers.
   (c) **Initialize** the set $S_j = \{\ \}, j = 1, \ldots, k$.
2. **Repeat**
   (a) **For** $i = 1, \ldots, M$ **do**
      i. **Set** $j^* = \text{argmin}_{m=1}^{k}\{D(x_i, c_m)\}$. In this formulation, j^* denotes the index of the closest center to the input sample $\overrightarrow{x_i}$, whereas $D(x, y)$ corresponds to the Euclidean distance function.
      ii. **Set** $S_{j^*} = S_{j^*} \cup \{x_i\}$.
   (b) **End For**
   (c) **For** $j = 1..k$ **do**
      i. **Calculate** the number of points in each cluster S_j and denote them as M_j.
      ii. **Calculate** every center $\overrightarrow{c_i}$ using the equation:
$$\overrightarrow{c_i} = \frac{1}{M_j}\sum_{x_i \in S_j} x_i$$
   (d) **End For**
3. **If** the centers $\overrightarrow{c_i}, j = 1, \ldots, k$ did not change then the loop is terminated.
4. **End Repeat**
5. The algorithm **returns** as output the estimated centers $\overrightarrow{c_i},\ i = 1, .., k$ and the associated variances $\sigma_i,\ i = 1, \ldots, k$

The set of parameters comprising the centers and variances of an RBF network with k hidden units is represented by a vector of dimension

$$n = k \times d + d \tag{4}$$

where d denotes the dimensionality of the input patterns. This vector has the following structure:

$$\overrightarrow{I} = \{c_{11}, c_{12}, c_{1d}, \sigma_1, c_{21}, c_{22}, \ldots, c_{2d}, \sigma_2, \ldots, c_{k1}, c_{k2}, \ldots, c_{kd}, \sigma_k\} \tag{5}$$

For this vector of parameters the bound vectors $\overrightarrow{L}$, $\overrightarrow{R}$ are constructed using the procedure depicted in Algorithm 2.

Algorithm 2 The algorithm that is used to construct the bound vectors $\overrightarrow{L}$, $\overrightarrow{R}$.

1. **Input**: the vector $\overrightarrow{I}$ of Equation 5.
2. **Input**: the scale factor used for the bound vectors denoted as F with $F \geq 1$.
3. **Input**: a small positive number v_e used as the low limit of the variables representing the variances of the RBF network.
4. **Set** $m = 1$
5. **For** $i = 1, \ldots, k$ **do**
   (a) **For** $j = 1, \ldots, d$ **do** //this loop is for the variables of each center
      i. **Set** $L_m = -F \times I_m$
      ii. **Set** $R_m = F \times I_m$
      iii. **Set** $m = m + 1$
   (b) **End For**
   (c) **Set** $L_m = v_e$ //set the bounds for the variables representing the variances.
   (d) **Set** $R_m = F \times I_m$
   (e) **Set** $m = m + 1$
6. **End For**
7. **Output**: The bound vectors $\overrightarrow{L}$, $\overrightarrow{R}$

2.2. The description of the second phase

In the second phase of the proposed algorithm, a genetic algorithm is employed to optimize the centers and variances of the RBF network within the search intervals defined by the vectors $\overrightarrow{L}$, $\overrightarrow{R}$ obtained during the first phase. The output weight vector $\overrightarrow{w}$ of the RBF network is subsequently computed by solving the corresponding linear system of equations. Within the proposed genetic algorithm, each chromosome follows the structure of the parameter vector $\overrightarrow{I}$ introduced in Equation 5, where all decision variables corresponding to the RBF centers and variances are encoded.

A suitably modified genetic algorithm was adopted in this stage because genetic algorithms constitute robust evolutionary optimization techniques that have demonstrated their effectiveness across a broad spectrum of practical applications. In addition, they are particularly well suited for integration with machine learning models, as they generally require only the formulation of a suitable fitness function while treating the underlying model as a black-box optimization problem. For example they have been applied to energy problems [52], water distribution problems [53], economic problems [54], training of of neural networks [55] etc. The steps of the used genetic algorithm have as follows:

1. **Initialization step**.
   (a) The algorithm has the following set of parameters:
      i. N_c, the desired number of chromosomes.
      ii. N_g, the maximum number of allowed generations.
      iii. p_s, the used selection rate with $p_s \leq 1$.
      iv. p_m, the used mutation rate with $p_m \leq 1$.
   (b) **Construct** a set of chromosomes with dimension n as defined in Equation 4. Each chromosome represents the centers and variances of any given RBF network and the value of each chromosome is initialized inside the bound vectors $\overrightarrow{L}$, $\overrightarrow{R}$.
   (c) **Set** $k = 0$, which represents the current number of generations.
2. **Fitness calculation step**.
   (a) **For** $i = 1, \ldots, N_c$ **do**

i. **Produce** an RBF network $R_i = R(\overrightarrow{x}, \overrightarrow{g_i})$. The centers and variances of R_i are derived from chromosome $\overrightarrow{g_i}$. The weight vector $\overrightarrow{w} = (w_1, w_2, \ldots, w_k)$ is calculated as follows:

A. **Set** $W = w_{kj}$

B. **Set** $\Phi = \phi_j(x_i)$, that denotes the outputs of the computation units.

C. **Set** $T = \{t_i = y_i, i = 1, .., M\}$, that denotes the real outputs for the training set.

D. **Solve** the following system of equations:

$$\Phi^T \left(T - \Phi W^T \right) = 0 \tag{6}$$

The solution is:

$$W^T = \left(\Phi^T \Phi \right)^{-1} \Phi^T T = \Phi^{\dagger} T \tag{7}$$

ii. **Calculate** the corresponding fitness value f_i as

$$f_i = \sum_{j=1}^{M} \left(R(\overrightarrow{x}_j, \overrightarrow{g_i}) - y_j \right)^2 \tag{8}$$

(b) **End For**

3. **Genetic operations step**.

(a) **Selection** procedure: The population is first ordered based on fitness. A fraction p_s of the population, corresponding to the top $p_s \times N_c$ chromosomes, is retained intact for the next generation. The rest of the population is regenerated by applying crossover and mutation operations to selected parent chromosomes.

(b) **Crossover** procedure: In this step $(1 - p_s) \times N_c$ new offsprings will be produced. For each pair $(\tilde{z}, \tilde{w})$ of these new chromosomes, two chromosomes (z, w) are chosen from the current population with the assistance of tournament selection. The new chromosomes will be created using the following equations:

$$\begin{aligned} \tilde{z}_i &= a_i z_i + (1 - a_i) w_i \\ \tilde{w}_i &= a_i w_i + (1 - a_i) z_i \end{aligned} \tag{9}$$

In this equation the a_i denote random numbers in the range $[-0.5, 1.5]$ [56].

(c) **Mutation** procedure: Select a random number $r \in [0, 1]$ for each element $\overrightarrow{g_{i,j}}, j = 1, \ldots, n$ for the corresponding chromosome g_i . If $r \leq p_m$, then this element is changed using the following scheme:

$$\overrightarrow{g_{i,j}} = \overrightarrow{g_{i,j}} + t \times p_c \times \overrightarrow{g_{i,j}} \tag{10}$$

where p_c is a number in range $[0, 1]$ that denotes the magnitude of changes applied to the current element and t is randomly selected integer that could be -1 or 1.

4. **Termination check step**.

(a) **Set** $k = k + 1$

(b) **If** $k < N_g$then **goto** Fitness Calculation step.

(c) **Otherwise,** obtain the best chromosome $\overrightarrow{g^*}$ and create the associated RBF network $R^* = R\left(\overrightarrow{x}, \overrightarrow{g^*}\right)$. The weight vector for this RBF network is also obtained by solving a similar system of equations as in Fitness Calculation Step.

(d) **Apply** a local search procedure to minimize the error function:

$$f^* = \sum_{j=1}^{M} \left(R^* \left(\overrightarrow{x}_j, \overrightarrow{g^*} \right) - y_j \right)^2 \quad (11)$$

A modified BFGS algorithm based on the method of Powell [57] was adopted as the local optimization procedure. This variant efficiently enforces the parameter bounds established during the preceding stage of the proposed methodology.

The overall structure of the second phase is illustrated in Figure 2. The procedure begins with the initialization of a population of chromosomes within the parameter bounds established during the first phase. The centers and variances of each candidate RBF network are extracted directly from the corresponding chromosome, the network weights are subsequently estimated by solving the corresponding linear system of equations, and a fitness value is assigned to each candidate based on the resulting prediction error. Following the evaluation of all candidates, the genetic operations of selection, crossover, and mutation are applied to produce the next generation. The generation counter is then incremented and checked against the maximum allowed number of generations. This cycle repeats until the termination criterion is met. At that point, the best chromosome identified throughout the search is used to reconstruct the final RBF network, whose weights are re-estimated by solving the linear system once more, after which a local optimization step based on the BFGS method is applied to further refine the solution.

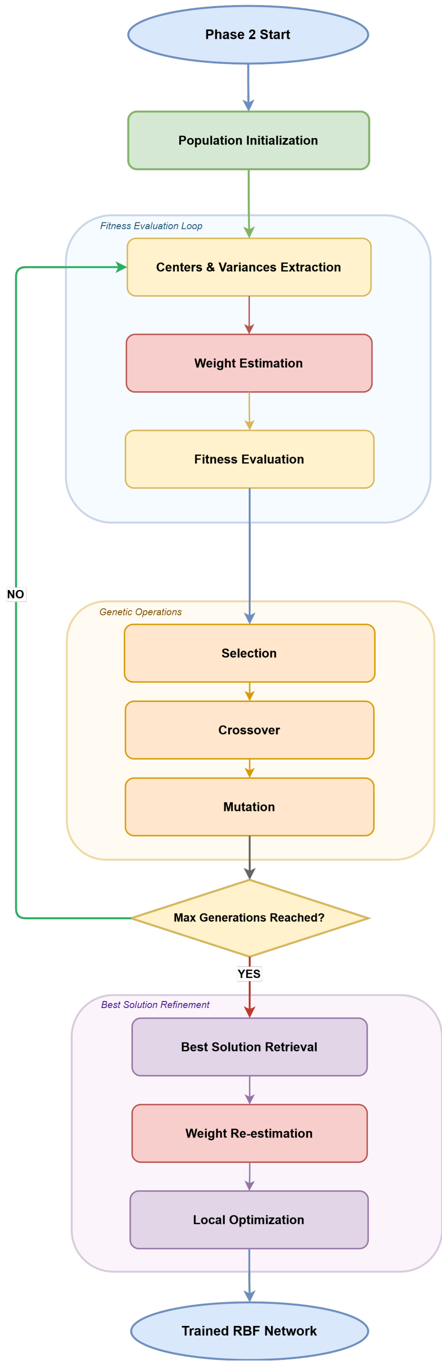

Figure 2. Flowchart of the second phase of the proposed training procedure.

3. Results

3.1. Experimental datasets The datasets used in the conducted experiments were obtained from the following online databases:

1. The UCI website, https://archive.ics.uci.edu/(accessed on 24 June 2026)[58]
2. The Keel website, https://sci2s.ugr.es/keel/datasets.php(accessed on 24 June 2026)[59].
3. The Statlib URL ftp://lib.stat.cmu.edu/datasets/index.html(accessed on 24 June 2026).

The list of classification datasets has as follows:

1. **Appendictis** dataset, which is a medical dataset[60].
2. **Alcohol** dataset, related to alcohol consumption and appeared in [61].
3. **Australian** dataset [62] used in economics.

4. **Balance** dataset [63], related to various psychological experiments.
5. **Bands** dataset, used to detect printing problems [64].
6. **Circular** dataset, which is produced artificially and contains two distinct classes.
7. **Cleveland** dataset, related to the presence of heart disease [65,66].
8. **Dermatology** dataset [67], a medical dataset.
9. **Ecoli**, used to detect protein problems [68].
10. The **Fert** dataset, related to sperm concentration problems.
11. **Glass** dataset, related to glass issues.
12. **Haberman**, a medical dataset that contains data used in the prediction of breast cancer.
13. **Hayes roth** dataset[69].
14. **Heart** dataset [70], which was used to detect heart diseases.
15. **HeartAttack**, a medical dataset related to heart diseases
16. **HouseVotes** dataset [71].
17. **Ionosphere** dataset, that was used to classify radar data from the ionosphere [72,73].
18. **Liverdisorder** dataset [74], which is a medical dataset.
19. **Lymography** dataset[75].
20. **Mammographic** dataset [76], which is a medical dataset related to breast tumors.
21. **Parkinsons** dataset, which is a medical dataset used to detect the presence of the Parkinson's Disease[77].
22. **Page Blocks** dataset, related to the determination of blocks inside various documents.
23. **Phoneme**, a dataset related to sound.
24. **Pima** dataset, which is a medical dataset used widely [78].
25. **Popfailures** dataset [79], used in various experiments regarding climate.
26. **Ring**, a dataset related to some distributions.
27. **Spiral** dataset, which was produced artificially and contains two classes.
28. **Regions2** dataset [80].
29. **Saheart** dataset [81], a medical dataset related to the presence of heart diseases.
30. **Satimage** dataset, related to satellite images.
31. **Segment** dataset [82], which is an image processing dataset.
32. The **Sonar** dataset [83].
33. **Spambase** dataset, regarding the detection of spam emails [84].
34. **Statheart**, which is medical dataset on heart diseases.
35. **Student**, which contains data from various experiments in schools [85].
36. **Transfusion** dataset [86].
37. **Wdbc** dataset [87], which is a medical dataset related the presence of breast tumors.
38. **Wine** dataset, used for the classification of the quality of wines [88,89].
39. **Eeg** dataset, which is an EEG dataset [90] . From this dataset the following cases were used: Z_F_S, ZONF_S and ZO_NF_S.
40. **Zoo** dataset [91], which was used to estimate the class of various animals.

In addition, the list of the used regression datasets has as follows:

1. **Abalone**, used to detect the age of abalones [92].
2. **Airfoil**, a dataset derived from NASA [93].
3. **Auto**, used to estimate the fuel consumption of cars.
4. **BK**, used to estimate the points scored in basketball games.
5. **BL**, that contains measurements from electricity experiments.
6. **Baseball**, used for the estimation of income of baseball players.
7. **Concrete**, a dataset that contains data from civil engineering [95].
8. **DEE**, a dataset related to the prediction of the prices of electricity.
9. **FY,** that contains measurements related to fruit flies.
10. **HO**, which was downloaded from the STATLIB repository.

11. **Housing**, a dataset that contains data from prices of houses [96].
12. **Laser**, a dataset containing data from various physics experiments.
13. **LW**, a dataset used to detect problems in the weight of babes.
14. **MB** dataset [97].
15. **Mortgage**, a dataset that contains economic data.
16. **NT** dataset[98].
17. **PL** dataset, which was downloaded from the STATLIB repository.
18. **Plastic**, a dataset that contains measurements from plastic experiments.
19. **PY** dataset[99].
20. **Quake**, related to measurements on quakes.
21. **SN**, which is a benchmark dataset obtained from the STATLIB repository.
22. **Stock**, related to the prices of stocks.
23. **Treasury**, which is a dataset contains economic data.
24. The **VE** dataset, which was downloaded from the STATLIB repository.

3.2. Experimental details

The RBF network implemented in this study was developed in ANSI C++ with the support of the open-source Armadillo library [102]. The optimization algorithms used were obtained from the OPTIMUS environment, downloaded from https://github.com/itsoulos/OPTIMUS/(accessed on 24 June 2026)[101]. Furthermore, derivative computations required by the optimization procedures were performed using the Adept automatic differentiation library [103]. To assess the performance and generalization capability of the proposed approach, a 10-fold cross-validation protocol was applied to all datasets. Each experimental setup was repeated 30 independent times, with a distinct initialization seed assigned to the random number generator in every run. Random number generation was carried out using the drand48() function provided by the C programming language. The average classification error is defined as:

$$E_C\left(N\left(\overrightarrow{x},\overrightarrow{w}\right)\right) = 100 \times \frac{\sum_{i=1}^{N}\left(\text{class}\left(N\left(\overrightarrow{x_i},\overrightarrow{w}\right)\right) - y_i\right)}{N} \tag{12}$$

The set T denotes the corresponding test set, with $T = (x_i, y_i),\ i = 1, \ldots, N$. Also, the average regression error is defined as follows:

$$E_R\left(N\left(\overrightarrow{x},\overrightarrow{w}\right)\right) = \frac{\sum_{i=1}^{N}\left(N\left(\overrightarrow{x_i},\overrightarrow{w}\right) - y_i\right)^2}{N} \tag{13}$$

The settings used in the conducted experiments are listed in Table 1.

Table 1. Experimental settings.

PARAMETER	MEANING	VALUE
N_c	Number of chromosomes	500
N_g	Maximum number of generations	500
p_s	Selection rate	0.10
p_m	Mutation rate	0.05
p_c	Mutation magnitude	0.05
F	Scale factor	4.0
v_e	Small value used in variances	0.01
k	Number of weights	10

Also, in the tables containing experimental results the following notation is used:

1. The column DATASET represents the name of the used dataset.
2. The column BFGS stands for the results obtained when the BFGS optimization procedure [104] is used to train an artificial neural network [105,106] with 10 processing nodes.
3. The ADAM column stands for the usage of the ADAM optimizer [107,108] to train a neural network with 10 processing nodes.
4. The column RBF represents the results where the traditional training method is used to train an RBF network with 10 processing nodes.
5. The column NEAT [109] is used to denote the results from NEAT (NeuroEvolution of Augmenting Topologies).
6. The column GRBF is used for the experimental results of the method presented in [110] to train an RBF network with 10 processing nodes.
7. The column PROPOSED stands for the results obtained by the incorporation of the proposed method.
8. The row labeled AVERAGE presents the mean classification or regression error obtained across the entire collection of datasets.

3.3. Comparison with other machine learning methods

The results of the comparative evaluation are presented in Table 2 for the classification problems and in Table 3 for the regression problems, while Figures 3 and 4 display the distribution of errors via box plots and report the outcomes of the overall Friedman test. For classification tasks (Table 2), the mean error rates were 35.18% for ADAM, 35.50% for GRBF, 33.25% for BFGS, 32.16% for NEAT, 28.68% for RBF, and 21.58% for the Proposed method. The overall Friedman test (Figure 3) yielded $p = 4.5 \times 10^{-10}$, confirming that the differences among methods are statistically significant at a very high confidence level. The Proposed method achieves the lowest mean error and outperforms the second-best method, RBF, by approximately 24.8%. At the individual dataset level, the Proposed method ranked first in 26 out of 42 datasets (61.9%), with particularly strong performance on RING, SPAMBASE, Z_F_S, and ZONF_S, as detailed in Table 2.

Table 2. Experimental results on the classification datasets, using a variety of machine learning methods.

DATASET	ADAM	BFGS	NEAT	RBF	GRBF	PROPOSED
APPENDICITIS	16.50%	18.00%	17.20%	12.23%	16.83%	17.80%
ALCOHOL	57.78%	41.50%	66.80%	49.38%	52.45%	20.43%
AUSTRALIAN	35.65%	38.13%	31.98%	34.89%	41.79%	18.90%
BALANCE	12.27%	8.64%	23.14%	33.53%	38.02%	17.50%
BANDS	36.92%	36.67%	34.30%	37.17%	43.95%	36.28%
CIRCULAR	19.95%	6.08%	35.18%	5.98%	21.43%	5.04%
CLEVELAND	67.55%	77.55%	53.44%	67.10%	67.47%	50.14%
DERMATOLOGY	26.14%	52.92%	32.43%	62.34%	61.46%	39.80%
ECOLI	64.43%	69.52%	43.44%	59.48%	70.58%	58.36%
FERT	23.98%	23.20%	15.37%	15.00%	15.00%	21.40%
GLASS	61.38%	54.67%	55.71%	50.05%	66.91%	52.76%
HABERMAN	29.00%	29.34%	24.04%	25.10%	24.37%	27.37%
HAYES ROTH	59.70%	37.33%	50.15%	64.36%	63.46%	40.08%
HEART	38.53%	39.44%	39.27%	31.20%	28.44%	22.00%
HEARTATTACK	45.55%	46.67%	32.34%	29.00%	40.48%	19.13%
HOUSEVOTES	7.48%	7.13%	10.89%	6.13%	11.99%	4.61%
IONOSPHERE	16.64%	15.29%	19.67%	16.22%	19.83%	12.83%
LIVERDISORDER	41.53%	42.59%	30.67%	30.84%	36.97%	30.41%
LYMOGRAPHY	39.79%	35.43%	33.70%	25.50%	29.33%	24.57%
MAMMOGRAPHIC	46.25%	17.24%	22.85%	21.38%	30.41%	17.72%
PARKINSONS	24.06%	27.58%	18.56%	17.41%	33.81%	15.53%
PAGE BLOCKS	34.27%	8.49%	10.22%	10.09%	10.83%	9.47%
PHONEME	29.43%	15.58%	22.34%	23.32%	27.70%	15.59%
PIMA	34.85%	35.59%	34.51%	25.78%	27.83%	23.59%
POPFAILURES	5.18%	5.24%	7.05%	7.04%	7.08%	5.61%
REGIONS2	29.85%	36.28%	33.23%	38.29%	39.98%	25.71%
RING	28.80%	29.44%	30.85%	21.67%	52.03%	5.05%
SAHEART	34.04%	37.48%	34.51%	32.19%	33.90%	28.89%
SATIMAGE	73.22%	71.73%	76.19%	40.19%	61.20%	32.97%
SEGMENT	49.75%	68.97%	66.72%	59.68%	54.25%	32.47%
SONAR	30.33%	25.85%	34.10%	27.90%	37.13%	24.00%
SPAMBASE	48.05%	18.16%	35.77%	29.35%	35.36%	10.54%
SPIRAL	47.67%	47.99%	48.66%	44.87%	50.02%	41.03%
STATHEART	44.04%	39.65%	44.36%	31.36%	42.94%	23.89%
STUDENT	5.13%	7.14%	10.20%	5.49%	33.26%	12.43%
TRANSFUSION	25.68%	25.84%	24.87%	26.41%	25.67%	25.23%
WDBC	35.35%	29.91%	12.88%	7.27%	8.82%	4.19%
WINE	29.40%	59.71%	25.43%	31.41%	31.47%	10.59%
Z_F_S	47.81%	39.37%	38.41%	13.16%	23.37%	4.68%
ZO_NF_S	47.43%	43.04%	43.75%	9.02%	22.18%	5.47%
ZONF_S	11.99%	15.62%	5.44%	4.03%	17.41%	1.82%
ZOO	14.13%	10.70%	20.27%	21.93%	33.50%	10.60%
AVERAGE	**35.18%**	**33.25%**	**32.16%**	**28.68%**	**35.50%**	**21.58%**

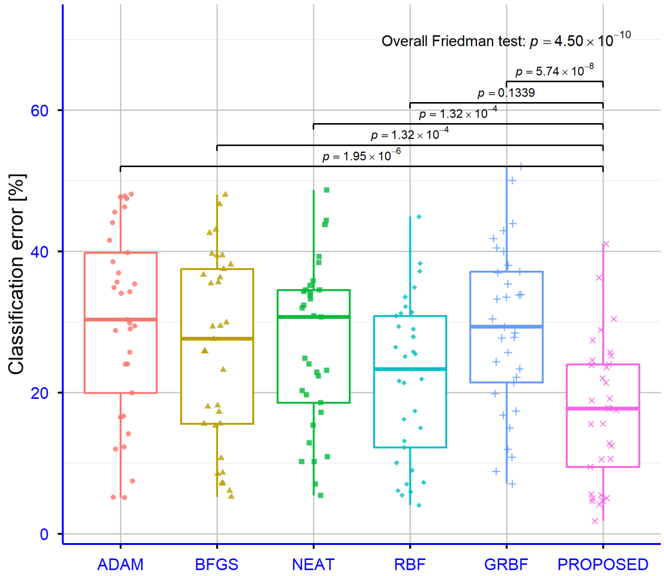

Figure 3. Classification error distribution across the six evaluated methods (overall Friedman test: $p = 4.5 \times 10^{-10}$)

Table 3. Experimental results on the used regression datasets using the series of machine learning methods.

DATASET	ADAM	BFGS	NEAT	RBF	GRBF	PROPOSED
ABALONE	4.30	5.69	9.88	7.37	9.98	6.37
AIRFOIL	0.005	0.003	0.067	0.27	0.121	0.004
AUTO	70.84	60.97	56.06	17.87	16.78	10.02
BK	0.025	0.28	0.15	0.02	0.023	0.021
BL	0.622	2.55	0.05	0.013	0.005	0.0005
BASEBALL	77.90	119.63	100.39	93.02	98.91	53.20
CONCRETE	0.078	0.066	0.081	0.011	0.015	0.008
DEE	0.63	2.36	1.512	0.17	0.25	0.16
FA	0.048	0.426	0.19	0.015	0.15	0.019
FY	0.038	0.22	0.08	0.011	0.041	0.051
HO	0.035	0.62	0.169	0.03	0.076	0.007
HOUSING	80.998	97.38	56.49	57.68	95.69	10.36
LASER	0.03	0.015	0.084	0.03	0.075	0.0198
LW	0.028	2.98	0.17	0.03	0.057	0.016
MB	0.06	0.129	0.061	5.43	0.159	0.066
MORTGAGE	9.24	8.23	14.11	1.45	1.92	0.0096
NT	0.006	0.129	0.027	13.97	0.01	0.006
PLASTIC	11.71	20.32	20.77	8.62	25.91	2.30
PY	0.321	0.578	0.075	0.012	0.029	0.015
PL	0.117	0.290	0.098	2.118	0.155	0.024
QUAKE	0.039	0.012	0.102	0.01	0.79	0.036
SN	0.026	0.400	0.174	0.027	0.027	0.024
STOCK	180.893	302.43	215.818	12.23	25.18	1.208
TREASURY	11.16	9.91	15.52	2.02	1.89	0.047
VE	0.359	1.92	0.045	0.024	0.023	0.04
AVERAGE	**17.98**	**25.50**	**19.69**	**8.90**	**11.13**	**3.36**

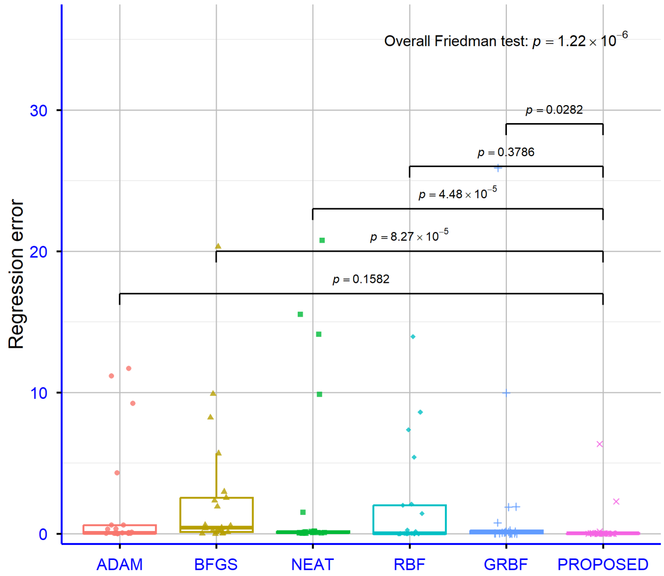

Figure 4. Regression error distribution for all evaluated methods (overall Friedman test: $p = 1.22 \times 10^{-6}$).

For regression tasks (Table 3), the mean error values were 17.98 for ADAM, 25.50 for BFGS, 19.69 for NEAT, 8.90 for RBF, 11.13 for GRBF, and 3.36 for the Proposed method. The overall Friedman test (Figure 4) yielded $p = 1.22 \times 10^{-6}$, again confirming statistically significant differences across methods. The Proposed method achieves a mean error of 3.36, representing nearly a threefold improvement over RBF (8.90) and tenfold over BFGS (25.50). Particularly notable results are observed on the STOCK dataset (1.208 versus 180.893 for ADAM), HOUSING (10.36 versus 97.38 for BFGS), and MORTGAGE (0.0096 versus 9.24 for ADAM), as shown in Table 3.

3.4. Experiments with the scale parameter F

Table 4 and Figure 5 present the effect of the scale parameter F on classification performance, while Table 5 and Figure 6 present the corresponding results for regression tasks. For classification (Table 4), the mean error rates were 25.93% for F=1, 22.85% for F=2, 21.58% for F=4, and 21.24% for F=8. The overall Friedman test (Figure 5) yielded $p = 9.36 \times 10^{-5}$, confirming that the choice of F has a statistically significant impact on performance. A clear decreasing trend in error is observed as F increases, with F=8 yielding a marginally lower average than F=4. However, the difference between these two settings is minimal (21.58% versus 21.24%), suggesting that F=4 represents a practical default with a favorable performance-to-complexity trade-off. For regression (Table 5), the mean error values were 4.69 for F=1, 4.03 for F=2, 3.36 for F=4, and 3.29 for F=8, with the Friedman test (Figure 6) yielding $p = 9.06 \times 10^{-4}$. A consistent improvement is again observed with increasing F, and the results for F=4 and F=8 are closely comparable across regression datasets.

Table 4. Experimental results on the classification datasets, using the proposed method and different values for scale parameter F.

DATASET	$F = 1$	$F = 2$	$F = 4$	$F = 8$
APPENDICITIS	17.60%	14.20%	17.80%	18.50%
ALCOHOL	30.09%	23.83%	20.43%	20.21%
AUSTRALIAN	16.35%	18.64%	18.90%	20.13%
BALANCE	34.97%	23.28%	17.50%	18.27%
BANDS	34.33%	35.03%	36.28%	34.14%
CIRCULAR	34.83%	8.55%	5.04%	3.76%
CLEVELAND	48.38%	49.03%	50.14%	50.28%
DERMATOLOGY	60.10%	38.57%	39.80%	38.26%
ECOLI	56.73%	59.36%	58.36%	59.61%
FERT	25.80%	22.80%	21.40%	21.70%
GLASS	53.90%	53.65%	52.76%	51.29%
HABERMAN	28.23%	27.24%	27.37%	27.13%
HAYES ROTH	59.69%	53.92%	40.08%	37.31%
HEART	22.67%	24.15%	22.00%	20.85%
HEARTATTACK	20.03%	19.67%	19.13%	19.17%
HOUSEVOTES	5.52%	5.26%	4.61%	5.00%
IONOSPHERE	11.20%	13.51%	12.83%	12.43%
LIVERDISORDER	30.85%	29.65%	30.41%	29.56%
LYMOGRAPHY	32.86%	20.50%	24.57%	23.26%
MAMMOGRAPHIC	28.14%	21.88%	17.72%	17.80%
PARKINSONS	17.84%	17.32%	15.53%	15.37%
PAGE BLOCKS	9.48%	9.43%	9.47%	9.36%
PHONEME	23.37%	20.32%	15.59%	15.82%
PIMA	23.91%	24.87%	23.59%	23.71%
POPFAILURES	10.18%	6.43%	5.61%	5.63%
REGIONS2	25.73%	25.97%	25.71%	25.65%
RING	5.55%	5.45%	5.05%	5.57%
SAHEART	29.00%	29.13%	28.89%	28.78%
SATIMAGE	33.62%	34.01%	32.97%	34.68%
SEGMENT	32.05%	31.62%	32.47%	33.33%
SONAR	44.05%	29.45%	24.00%	22.05%
SPAMBASE	13.39%	10.84%	10.54%	10.44%
SPIRAL	36.80%	41.27%	41.03%	41.49%
STATHEART	22.74%	25.22%	23.89%	23.26%
STUDENT	25.50%	19.33%	12.43%	4.25%
TRANSFUSION	25.19%	24.92%	25.23%	24.65%
WDBC	4.86%	5.19%	4.19%	4.37%
WINE	11.29%	9.82%	10.59%	10.41%
Z_F_S	5.03%	6.80%	4.68%	5.37%
ZO_NF_S	5.67%	5.74%	5.47%	7.22%
ZONF_S	2.45%	2.22%	1.82%	2.30%
ZOO	29.00%	11.80%	10.60%	9.50%
AVERAGE	**25.93%**	**22.85%**	**21.58%**	**21.24%**

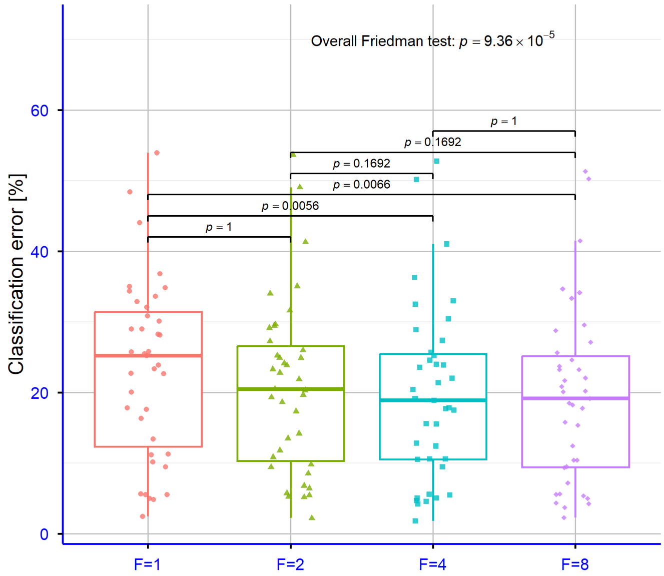

Figure 5. Effect of the scale parameter F on classification accuracy of the Proposed method (overall Friedman test: $p = 9.36 \times 10^{-5}$).

Table 5. Experimental results on the used regression datasets using the proposed method and a series of values for the parameter F.

DATASET	$F = 1$	$F = 2$	$F = 4$	$F = 8$
ABALONE	10.32	7.76	6.37	5.83
AIRFOIL	0.004	0.004	0.004	0.004
AUTO	10.06	9.68	10.02	9.53
BK	0.025	0.02	0.021	0.003
BL	0.0098	0.0052	0.0005	0.041
BASEBALL	67.54	59.65	53.20	51.28
CONCRETE	0.027	0.022	0.008	0.005
DEE	0.16	0.16	0.16	0.16
FA	0.02	0.019	0.019	0.017
FY	0.046	0.047	0.051	0.053
HO	0.007	0.007	0.007	0.007
HOUSING	12.43	10.45	10.36	11.70
LASER	0.096	0.044	0.0198	0.009
LW	0.032	0.028	0.016	0.014
MB	0.067	0.066	0.066	0.06
MORTGAGE	0.011	0.009	0.0096	0.011
NT	0.006	0.006	0.006	0.006
PLASTIC	11.74	10.66	2.30	2.24
PY	0.029	0.017	0.015	0.014
PL	0.117	0.106	0.024	0.022
QUAKE	0.036	0.036	0.036	0.036
SN	0.031	0.025	0.024	0.023
STOCK	4.28	1.92	1.208	1.08
TREASURY	0.041	0.044	0.047	0.052
VE	0.026	0.034	0.04	0.035
AVERAGE	**4.69**	**4.03**	**3.36**	**3.29**

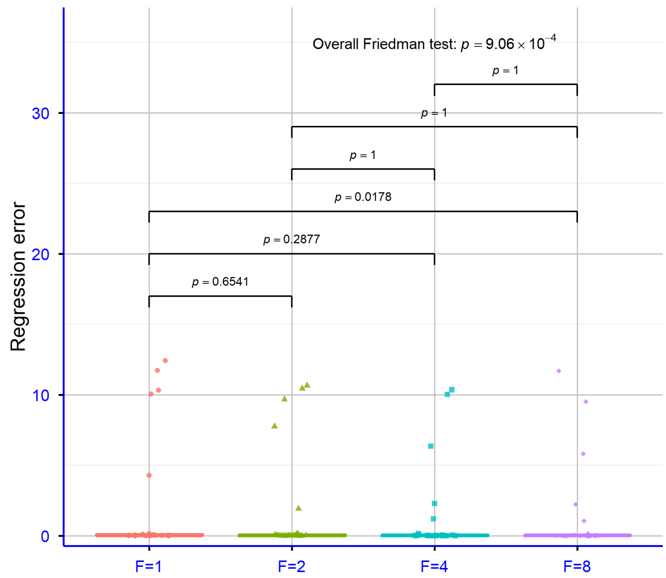

Figure 6. Sensitivity of regression performance to the scale parameter F (overall Friedman test: $p = 9.06 \times 10^{-4}$).

3.5. *Experiments with number of chromosomes N_c* 327

Table 6 and Figure 7 illustrate the effect of the chromosome population size N_c on classification performance, while Table 7 and Figure 8 present the corresponding regression results. For classification (Table 6), the mean error rates were 23.03% for N_c=100, 22.44% for N_c=200, 21.58% for N_c=500, and 21.55% for N_c=1000. The overall Friedman test (Figure 7) yielded p = 0.0037, confirming a statistically significant effect of N_c on classification accuracy. A steady improvement is observed from N_c=100 to N_c=500, while the increase to N_c=1000 yields no meaningful additional gain, suggesting that a population of 500 chromosomes offers a well-balanced trade-off between computational cost and predictive performance. For regression (Table 7), the mean error values were 3.63 for N_c=100, 3.45 for N_c=200, 3.36 for N_c=500, and 3.55 for N_c=1000. The overall Friedman test (Figure 8) yielded $p = 0.2415$, a value that does not reach statistical significance, indicating that for regression tasks the chromosome count does not substantially influence performance and that N_c=500 remains a sufficient and well-justified setting.

Table 6. Experimental results on the classification datasets, using the proposed method and different values the number of chromsosomes denoted as N_c.

DATASET	$N_c = 100$	$N_c = 200$	$N_c = 500$	$N_c = 1000$
APPENDICITIS	20.00%	19.40%	17.80%	15.40%
ALCOHOL	25.32%	20.21%	20.43%	19.36%
AUSTRALIAN	23.87%	20.94%	18.90%	17.81%
BALANCE	17.18%	16.97%	17.50%	17.84%
BANDS	38.28%	36.61%	36.28%	36.17%
CIRCULAR	11.10%	6.42%	5.04%	5.29%
CLEVELAND	48.66%	48.86%	50.14%	50.38%
DERMATOLOGY	41.69%	41.94%	39.80%	40.83%
ECOLI	60.40%	60.76%	58.36%	59.52%
FERT	22.50%	21.10%	21.40%	24.00%
GLASS	55.99%	55.14%	52.76%	51.62%
HABERMAN	26.17%	26.77%	27.37%	27.53%
HAYES ROTH	41.61%	42.39%	40.08%	38.08%
HEART	22.85%	23.34%	22.00%	21.59%
HEARTATTACK	17.90%	18.87%	19.13%	18.83%
HOUSEVOTES	6.65%	6.35%	4.61%	5.35%
IONOSPHERE	14.66%	14.12%	12.83%	11.83%
LIVERDISORDER	29.50%	29.59%	30.41%	29.77%
LYMOGRAPHY	29.72%	29.79%	24.57%	22.43%
MAMMOGRAPHIC	18.27%	17.77%	17.72%	17.84%
PARKINSONS	15.95%	15.68%	15.53%	16.05%
PAGE BLOCKS	9.40%	9.36%	9.47%	10.43%
PHONEME	16.00%	15.54%	15.59%	16.37%
PIMA	23.50%	23.66%	23.59%	23.23%
POPFAILURES	5.91%	6.33%	5.61%	5.46%
REGIONS2	25.71%	25.66%	25.71%	25.43%
RING	5.88%	5.88%	5.05%	4.46%
SAHEART	28.46%	28.00%	28.89%	28.94%
SATIMAGE	34.96%	35.21%	32.97%	35.15%
SEGMENT	33.05%	32.73%	32.47%	31.95%
SONAR	25.75%	25.00%	24.00%	22.95%
SPAMBASE	12.41%	11.74%	10.54%	10.11%
SPIRAL	40.59%	41.39%	41.03%	41.11%
STATHEART	23.78%	24.30%	23.89%	25.11%
STUDENT	22.18%	17.10%	12.43%	11.45%
TRANSFUSION	25.91%	25.74%	25.23%	25.34%
WDBC	4.54%	4.54%	4.19%	4.75%
WINE	11.00%	10.12%	10.59%	10.71%
Z_F_S	5.90%	5.43%	4.68%	6.07%
ZO_NF_S	7.74%	7.34%	5.47%	5.70%
ZONF_S	3.26%	2.52%	1.82%	2.60%
ZOO	13.10%	11.90%	10.60%	10.40%
AVERAGE	**23.03%**	**22.44%**	**21.58%**	**21.55%**

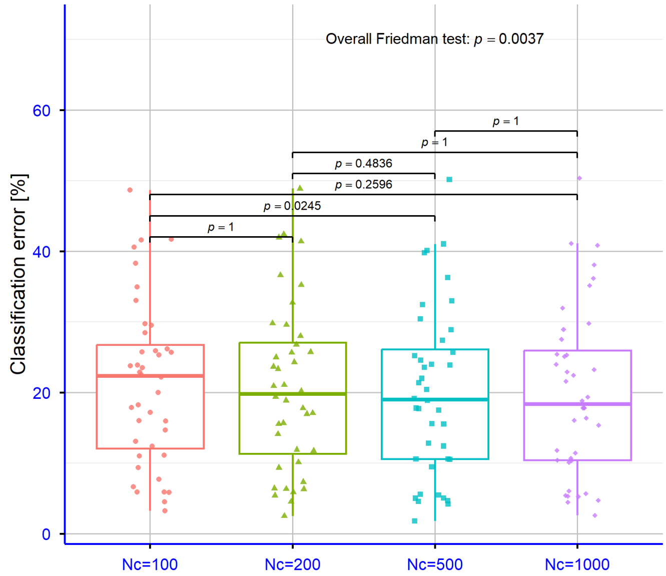

Figure 7. Impact of population size Nc on classification results of the Proposed method (overall Friedman test: $p = 0.0037$).

Table 7. Experimental results on the used regression datasets using the proposed method and different values for the number of chromosomes N_c.

DATASET	$N_c = 100$	$N_c = 200$	$N_c = 500$	$N_c = 1000$
ABALONE	7.24	6.90	6.37	6.13
AIRFOIL	0.004	0.004	0.004	0.004
AUTO	9.70	9.57	10.02	9.88
BK	0.02	0.02	0.021	0.022
BL	0.0007	0.0006	0.0005	0.022
BASEBALL	56.15	53.69	53.20	56.45
CONCRETE	0.01	0.009	0.008	0.007
DEE	0.23	0.16	0.16	0.16
FA	0.025	0.022	0.019	0.016
FY	0.049	0.049	0.051	0.051
HO	0.007	0.007	0.007	0.008
HOUSING	10.50	10.92	10.36	11.94
LASER	0.058	0.025	0.0198	0.017
LW	0.02	0.018	0.016	0.014
MB	0.06	0.063	0.066	0.065
MORTGAGE	0.014	0.008	0.0096	0.01
NT	0.006	0.006	0.006	0.006
PLASTIC	5.11	3.37	2.30	2.32
PY	0.015	0.016	0.015	0.013
PL	0.045	0.032	0.024	0.023
QUAKE	0.036	0.036	0.036	0.036
SN	0.023	0.023	0.024	0.024
STOCK	1.31	1.22	1.208	1.38
TREASURY	0.048	0.048	0.047	0.047
VE	0.039	0.042	0.04	0.047
AVERAGE	**3.63**	**3.45**	**3.36**	**3.55**

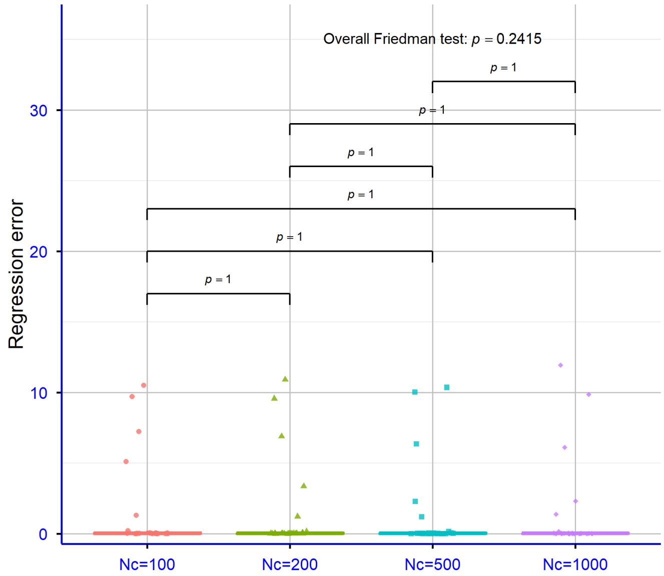

Figure 8. Regression error under varying chromosome population sizes Nc (overall Friedman test: $p = 0.2415$).

4. Conclusions

This paper introduced a two-phase evolutionary methodology for the effective training of Radial Basis Function networks. In the first phase, the K-means clustering algorithm is employed to establish informed search bounds for the network centers and variances, thereby constraining the subsequent optimization within a meaningful parameter space. In the second phase, a genetic algorithm operates within these bounds to identify optimal center and variance configurations, while the network weights are determined analytically by solving a system of linear equations. The proposed method was evaluated across a large and diverse collection of 42 classification and 25 regression benchmark datasets, consistently outperforming five competing approaches, namely ADAM, BFGS, NEAT, RBF, and GRBF. The mean classification error of 21.58% and the mean regression error of 3.36 represent substantial improvements over all baselines, and the differences were confirmed to be statistically significant by the Friedman test. The sensitivity analysis further demonstrated that the method is robust across a range of parameter settings, with a population size of 500 chromosomes and a scale factor of F=4 providing a reliable and computationally efficient default configuration.

Several directions are proposed for future investigation. First, the computational cost of the genetic algorithm could be reduced through parallelization, enabling the simultaneous evaluation of multiple candidate solutions across multi-core or distributed computing environments. Second, adaptive strategies for the automatic selection of key parameters, such as the population size N_c, the number of generations N_g, and the scale factor F, could further reduce the need for manual tuning while improving performance across diverse problem types. Third, the current Gaussian basis function could be replaced or complemented by alternative kernel functions, such as multiquadric or inverse multiquadric kernels, potentially broadening the class of problems the model can address effectively. Fourth, the integration of feature selection mechanisms within the training pipeline could improve both accuracy and interpretability, particularly for high-dimensional datasets. Finally, extending the proposed framework to deep or multi-hidden-layer RBF architectures represents a promising avenue for tackling more complex learning tasks.

Author Contributions: V.C. and I.G.T. conducted the experiments, employing several datasets and provided the comparative experiments. D.T. and V.C. performed the statistical analysis and prepared the manuscript. All authors have read and agreed to the published version of the manuscript.

Funding: This research has been financed by the European Union: Next Generation EU through the Program Greece 2.0 National Recovery and Resilience Plan, under the call RESEARCH—CREATE—INNOVATE, project name "iCREW: Intelligent small craft simulator for advanced crew training using Virtual Reality techniques" (project code:TAEDK-06195).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Raissi, M., & Karniadakis, G. E. (2018). Hidden physics models: Machine learning of nonlinear partial differential equations. Journal of Computational Physics, 357, 125-141.
2. Kashinath, K., Mustafa, M., Albert, A., Wu, J. L., Jiang, C., Esmaeilzadeh, S., ... & Prabhat, N. (2021). Physics-informed machine learning: case studies for weather and climate modelling. Philosophical Transactions of the Royal Society A, 379(2194), 20200093.
3. Viquar, M., Basak, S., Dasgupta, A., Agrawal, S., & Saha, S. (2019). Machine learning in astronomy: A case study in quasar-star classification. Emerging Technologies in Data Mining and Information Security: Proceedings of IEMIS 2018, Volume 3, 827-836.
4. Luo, S., Leung, A. P., Hui, C. Y., & Li, K. L. (2020). An investigation on the factors affecting machine learning classifications in gamma-ray astronomy. Monthly Notices of the Royal Astronomical Society, 492(4), 5377-5390.
5. Meuwly, M. (2021). Machine learning for chemical reactions. Chemical Reviews, 121(16), 10218-10239.
6. J.A. Aguiar, M.L. Gong, T.Tasdizen, Crystallographic prediction from diffraction and chemistry data for higher throughput classification using machine learning, Computational Materials Science **173**, 109409, 2020.
7. S.S. Yadav, S.M. Jadhav, Deep convolutional neural network based medical image classification for disease diagnosis. J Big Data **6**, 113, 2019.
8. L. Qing, W. Linhong , D. Xuehai, A Novel Neural Network-Based Method for Medical Text Classification, Future Internet **11**, 255, 2019.
9. Athey, S. (2018). The impact of machine learning on economics. In The economics of artificial intelligence: An agenda (pp. 507-547). University of Chicago Press.
10. R. Hafezi, J. Shahrabi, E. Hadavandi, A bat-neural network multi-agent system (BNNMAS) for stock price prediction: Case study of DAX stock price, Applied Soft Computing **29**, pp. 196-210, 2015.
11. Ghai, D., Tripathi, S. L., Saxena, S., Chanda, M., & Alazab, M. (Eds.). (2022). Machine learning algorithms for signal and image processing. John Wiley & Sons.
12. Ahmed, N. K., Atiya, A. F., Gayar, N. E., & El-Shishiny, H. (2010). An empirical comparison of machine learning models for time series forecasting. Econometric reviews, 29(5-6), 594-621.
13. Z. Huang, H. Chen, C.-Jung Hsu, W.-Hwa Chen, S. Wu, Credit rating analysis with support vector machines and neural networks: a market comparative study, Decision Support Systems **37**, pp. 543-558, 2004.
14. K. Peta, J. Żurek, Prediction of air leakage in heat exchangers for automotive applications using artificial neural networks, In: 2018 9th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York, NY, USA, pp. 721-725, 2018.
15. J. Park and I. W. Sandberg, Universal Approximation Using Radial-Basis-Function Networks, Neural Computation **3**, pp. 246-257, 1991.
16. G.A. Montazer, D. Giveki, M. Karami, H. Rastegar, Radial basis function neural networks: A review. Comput. Rev. J **1**, pp. 52-74, 2018.
17. P. Teng, Machine-learning quantum mechanics: Solving quantum mechanics problems using radial basis function networks, Phys. Rev. E **98**, 033305, 2018.
18. R. Jovanović, A. Sretenovic, Ensemble of radial basis neural networks with K-means clustering for heating energy consumption prediction, FME Transactions **45**, pp. 51-57, 2017.
19. V.I. Gorbachenko, M.V. Zhukov, Solving boundary value problems of mathematical physics using radial basis function networks. Comput. Math. and Math. Phys. **57**, pp. 145–155, 2017.

20. J. Määttä, V. Bazaliy, J. Kimari, F. Djurabekova, K. Nordlund, T. Roos, Gradient-based training and pruning of radial basis function networks with an application in materials physics, Neural Networks **133**, pp. 123-131, 2021.
21. Nam Mai-Duy, Thanh Tran-Cong, Numerical solution of differential equations using multiquadric radial basis function networks, Neural Networks 14, pp. 185-199, 2001.
22. N. Mai-Duy, Solving high order ordinary differential equations with radial basis function networks. Int. J. Numer. Meth. Engng. **62**, pp. 824-852, 2005.
23. S.A. Sarra, Adaptive radial basis function methods for time dependent partial differential equations, Applied Numerical Mathematics **54**, pp. 79-94, 2005.
24. R. -J. Lian, Adaptive Self-Organizing Fuzzy Sliding-Mode Radial Basis-Function Neural-Network Controller for Robotic Systems, IEEE Transactions on Industrial Electronics **61**, pp. 1493-1503, 2014.
25. M. Vijay, D. Jena, Backstepping terminal sliding mode control of robot manipulator using radial basis functional neural networks. Computers & Electrical Engineering **67**, pp. 690-707, 2018.
26. M.J. Er, S. Wu, J. Lu, H.L. Toh, Face recognition with radial basis function (RBF) neural networks, IEEE Transactions on Neural Networks **13**, pp. 697-710, 2002.
27. C. Laoudias, P. Kemppi and C. G. Panayiotou, Localization Using Radial Basis Function Networks and Signal Strength Fingerprints in WLAN, GLOBECOM 2009 - 2009 IEEE Global Telecommunications Conference, Honolulu, HI, 2009, pp. 1-6, 2009.
28. M. Azarbad, S. Hakimi, A. Ebrahimzadeh, Automatic recognition of digital communication signal, International journal of energy, information and communications **3**, pp. 21-33, 2012.
29. D.L. Yu, J.B. Gomm, D. Williams, Sensor fault diagnosis in a chemical process via RBF neural networks, Control Engineering Practice **7**, pp. 49-55, 1999.
30. V. Shankar, G.B. Wright, A.L. Fogelson, R.M. Kirby, A radial basis function (RBF) finite difference method for the simulation of reaction–diffusion equations on stationary platelets within the augmented forcing method, Int. J. Numer. Meth. Fluids **75**, pp. 1-22, 2014.
31. W. Shen, X. Guo, C. Wu, D. Wu, Forecasting stock indices using radial basis function neural networks optimized by artificial fish swarm algorithm, Knowledge-Based Systems 24, pp. 378-385, 2011.
32. J. A. Momoh, S. S. Reddy, Combined Economic and Emission Dispatch using Radial Basis Function, 2014 IEEE PES General Meeting | Conference & Exposition, National Harbor, MD, pp. 1-5, 2014.
33. P. Sohrabi, B. Jodeiri Shokri, H. Dehghani, Predicting coal price using time series methods and combination of radial basis function (RBF) neural network with time series. Miner Econ 2021.
34. U. Ravale, N. Marathe, P. Padiya, Feature Selection Based Hybrid Anomaly Intrusion Detection System Using K Means and RBF Kernel Function, Procedia Computer Science **45**, pp. 428-435, 2015.
35. M. Lopez-Martin, A. Sanchez-Esguevillas, J. I. Arribas, B. Carro, Network Intrusion Detection Based on Extended RBF Neural Network With Offline Reinforcement Learning, IEEE Access **9**, pp. 153153-153170, 2021.
36. Gan, M., Peng, H., & Dong, X. P. (2012). A hybrid algorithm to optimize RBF network architecture and parameters for nonlinear time series prediction. Applied Mathematical Modelling, 36(7), 2911-2919.
37. G. Sideratos and N. Hatziargyriou, "Using Radial Basis Neural Networks to Estimate Wind Power Production," 2007 IEEE Power Engineering Society General Meeting, Tampa, FL, USA, 2007, pp. 1-7, doi: 10.1109/PES.2007.385812.
38. L.I. Kuncheva, Initializing of an RBF network by a genetic algorithm, Neurocomputing **14**, pp. 273-288, 1997.
39. F. Ros, M. Pintore, A. Deman, J.R. Chrétien, Automatical initialization of RBF neural networks, Chemometrics and Intelligent Laboratory Systems **87**, pp. 26-32, 2007.
40. D. Wang, X.J. Zeng, J.A. Keane, A clustering algorithm for radial basis function neural network initialization, Neurocomputing **77**, pp. 144-155, 2012.
41. N. Benoudjit, M. Verleysen, On the Kernel Widths in Radial-Basis Function Networks, Neural Processing Letters **18**, pp. 139–154, 2003.
42. R. Neruda, P. Kudova, Learning methods for radial basis function networks, Future Generation Computer Systems **21**, pp. 1131–1142, 2005.
43. E. Ricci, R. Perfetti, Improved pruning strategy for radial basis function networks with dynamic decay adjustment, Neurocomputing **69**, pp. 1728-1732, 2006.
44. Guang-Bin Huang, P. Saratchandran and N. Sundararajan, A generalized growing and pruning RBF (GGAP-RBF) neural network for function approximation, IEEE Transactions on Neural Networks **16**, pp. 57-67, 2005.
45. M. Bortman and M. Aladjem, A Growing and Pruning Method for Radial Basis Function Networks, IEEE Transactions on Neural Networks **20**, pp. 1039-1045, 2009.
46. R. Yokota, L.A. Barba, M. G. Knepley, PetRBF — A parallel O(N) algorithm for radial basis function interpolation with Gaussians, Computer Methods in Applied Mechanics and Engineering **199**, pp. 1793-1804, 2010.

47. C. Lu, N. Ma, Z. Wang, Fault detection for hydraulic pump based on chaotic parallel RBF network, EURASIP J. Adv. Signal Process. **2011**, 49, 2011.
48. J. Park and I. W. Sandberg, "Universal Approximation Using Radial-Basis-Function Networks," in Neural Computation, vol. 3, no. 2, pp. 246-257, June 1991, doi: 10.1162/neco.1991.3.2.246.
49. E.U. Oti, M.O. Olusola, F.C. Eze, S.U. Enogwe, Comprehensive review of K-Means clustering algorithms, Criterion **12**, pp. 22-23, 2021.
50. D. Goldberg, Genetic Algorithms in Search, Optimization and Machine Learning, Addison-Wesley Publishing Company, Reading, Massachussets, 1989.
51. Z. Michaelewicz, Genetic Algorithms + Data Structures = Evolution Programs. Springer - Verlag, Berlin, 1996.
52. S.A. Grady, M.Y. Hussaini, M.M. Abdullah, Placement of wind turbines using genetic algorithms, Renewable Energy **30**, pp. 259-270, 2005.
53. Parvaze, S., Kumar, R., Khan, J. N., Al-Ansari, N., Parvaze, S., Vishwakarma, D. K., ... & Kuriqi, A. (2023). Optimization of water distribution systems using genetic algorithms: A review. Archives of Computational Methods in Engineering, 30(7), 4209-4244.
54. Gordini, N. (2014). A genetic algorithm approach for SMEs bankruptcy prediction: Empirical evidence from Italy. Expert systems with applications, 41(14), 6433-6445.
55. Ding, S., Su, C., & Yu, J. (2011). An optimizing BP neural network algorithm based on genetic algorithm. Artificial intelligence review, 36(2), 153-162.
56. P. Kaelo, M.M. Ali, Integrated crossover rules in real coded genetic algorithms, European Journal of Operational Research **176**, pp. 60-76, 2007.
57. M.J.D Powell, A Tolerant Algorithm for Linearly Constrained Optimization Calculations, Mathematical Programming **45**, pp. 547-566, 1989.
58. M. Kelly, R. Longjohn, K. Nottingham, The UCI Machine Learning Repository, https://archive.ics.uci.edu.
59. J. Alcalá-Fdez, A. Fernandez, J. Luengo, J. Derrac, S. García, L. Sánchez, F. Herrera. KEEL Data-Mining Software Tool: Data Set Repository, Integration of Algorithms and Experimental Analysis Framework. Journal of Multiple-Valued Logic and Soft Computing 17, pp. 255-287, 2011.
60. Weiss, Sholom M. and Kulikowski, Casimir A., Computer Systems That Learn: Classification and Prediction Methods from Statistics, Neural Nets, Machine Learning, and Expert Systems, Morgan Kaufmann Publishers Inc, 1991.
61. Tzimourta, K.D.; Tsoulos, I.; Bilero, I.T.; Tzallas, A.T.; Tsipouras, M.G.; Giannakeas, N. Direct Assessment of Alcohol Consumption in Mental State Using Brain Computer Interfaces and Grammatical Evolution. Inventions 2018, 3, 51.
62. J.R. Quinlan, Simplifying Decision Trees. International Journal of Man-Machine Studies **27**, pp. 221-234, 1987.
63. T. Shultz, D. Mareschal, W. Schmidt, Modeling Cognitive Development on Balance Scale Phenomena, Machine Learning **16**, pp. 59-88, 1994.
64. B. Evans, D. Fisher, Overcoming process delays with decision tree induction, IEEE Expert **9**, pp. 60-66, 1994.
65. Z.H. Zhou,Y. Jiang, NeC4.5: neural ensemble based C4.5," in IEEE Transactions on Knowledge and Data Engineering **16**, pp. 770-773, 2004.
66. R. Setiono , W.K. Leow, FERNN: An Algorithm for Fast Extraction of Rules from Neural Networks, Applied Intelligence **12**, pp. 15-25, 2000.
67. G. Demiroz, H.A. Govenir, N. Ilter, Learning Differential Diagnosis of Eryhemato-Squamous Diseases using Voting Feature Intervals, Artificial Intelligence in Medicine. **13**, pp. 147–165, 1998.
68. P. Horton, K.Nakai, A Probabilistic Classification System for Predicting the Cellular Localization Sites of Proteins, In: Proceedings of International Conference on Intelligent Systems for Molecular Biology **4**, pp. 109-15, 1996.
69. B. Hayes-Roth, B., F. Hayes-Roth. Concept learning and the recognition and classification of exemplars. Journal of Verbal Learning and Verbal Behavior **16**, pp. 321-338, 1977.
70. I. Kononenko, E. Šimec, M. Robnik-Šikonja, Overcoming the Myopia of Inductive Learning Algorithms with RELIEFF, Applied Intelligence **7**, pp. 39–55, 1997
71. R.M. French, N. Chater, Using noise to compute error surfaces in connectionist networks: a novel means of reducing catastrophic forgetting, Neural Comput. **14**, pp. 1755-1769, 2002.
72. J.G. Dy , C.E. Brodley, Feature Selection for Unsupervised Learning, The Journal of Machine Learning Research **5**, pp 845–889, 2004.
73. S. J. Perantonis, V. Virvilis, Input Feature Extraction for Multilayered Perceptrons Using Supervised Principal Component Analysis, Neural Processing Letters **10**, pp 243–252, 1999.
74. J. Garcke, M. Griebel, Classification with sparse grids using simplicial basis functions, Intell. Data Anal. **6**, pp. 483-502, 2002.
75. G. Cestnik, I. Konenenko, I. Bratko, Assistant-86: A Knowledge-Elicitation Tool for Sophisticated Users. In: Bratko, I. and Lavrac, N., Eds., Progress in Machine Learning, Sigma Press, Wilmslow, pp. 31-45, 1987.

76. M. Elter, R. Schulz-Wendtland, T. Wittenberg, The prediction of breast cancer biopsy outcomes using two CAD approaches that both emphasize an intelligible decision process, Med Phys. **34**, pp. 4164-72, 2007.
77. Little MA, McSharry PE, Hunter EJ, Spielman J, Ramig LO. Suitability of dysphonia measurements for telemonitoring of Parkinson's disease. IEEE Trans Biomed Eng. 2009;56(4):1015. doi:10.1109/TBME.2008.2005954
78. J.W. Smith, J.E. Everhart, W.C. Dickson, W.C. Knowler, R.S. Johannes, Using the ADAP learning algorithm to forecast the onset of diabetes mellitus, In: Proceedings of the Symposium on Computer Applications and Medical Care IEEE Computer Society Press, pp.261-265, 1988.
79. D.D. Lucas, R. Klein, J. Tannahill, D. Ivanova, S. Brandon, D. Domyancic, Y. Zhang, Failure analysis of parameter-induced simulation crashes in climate models, Geoscientific Model Development **6**, pp. 1157-1171, 2013.
80. Giannakeas, N., Tsipouras, M.G., Tzallas, A.T., Kyriakidi, K., Tsianou, Z.E., Manousou, P., Hall, A., Karvounis, E.C., Tsianos, V., Tsianos, E. A clustering based method for collagen proportional area extraction in liver biopsy images (2015) Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS, 2015-November, art. no. 7319047, pp. 3097-3100.
81. T. Hastie, R. Tibshirani, Non-parametric logistic and proportional odds regression, JRSS-C (Applied Statistics) **36**, pp. 260–276, 1987.
82. M. Dash, H. Liu, P. Scheuermann, K. L. Tan, Fast hierarchical clustering and its validation, Data & Knowledge Engineering **44**, pp 109–138, 2003.
83. Gorman, R.P.; Sejnowski, T.J. Analysis of Hidden Units in a Layered Network Trained to Classify Sonar Targets. Neural Netw. 1988, 1, 75–89.
84. Cranor, Lorrie F., LaMacchia, Brian A. Spam!, Communications of the ACM, 41(8):74-83, 1998.
85. P. Cortez, A. M. Gonçalves Silva, Using data mining to predict secondary school student performance, In Proceedings of 5th FUture BUsiness TEChnology Conference (FUBUTEC 2008) (pp. 5–12). EUROSIS-ETI, 2008.
86. I-Cheng Yeh, King-Jang Yang, Tao-Ming Ting, Knowledge discovery on RFM model using Bernoulli sequence, Expert Systems with Applications **36**, pp. 5866-5871, 2009.
87. W.H. Wolberg, O.L. Mangasarian, Multisurface method of pattern separation for medical diagnosis applied to breast cytology, Proc Natl Acad Sci U S A. **87**, pp. 9193–9196, 1990.
88. M. Raymer, T.E. Doom, L.A. Kuhn, W.F. Punch, Knowledge discovery in medical and biological datasets using a hybrid Bayes classifier/evolutionary algorithm. IEEE transactions on systems, man, and cybernetics. Part B, Cybernetics : a publication of the IEEE Systems, Man, and Cybernetics Society, **33** , pp. 802-813, 2003.
89. P. Zhong, M. Fukushima, Regularized nonsmooth Newton method for multi-class support vector machines, Optimization Methods and Software **22**, pp. 225-236, 2007.
90. R.G. Andrzejak, K. Lehnertz, F. Mormann, C. Rieke, P. David, and C. E. Elger, Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state, Phys. Rev. E **64**, pp. 1-8, 2001.
91. M. Koivisto, K. Sood, Exact Bayesian Structure Discovery in Bayesian Networks, The Journal of Machine Learning Research **5**, pp. 549–573, 2004.
92. W. J Nash, T.L. Sellers, S.R. Talbot, A.J. Cawthor, W.B. Ford, The Population Biology of Abalone (_Haliotis_ species) in Tasmania. I. Blacklip Abalone (_H. rubra_) from the North Coast and Islands of Bass Strait, Sea Fisheries Division, Technical Report No. 48 (ISSN 1034-3288), 1994.
93. T.F. Brooks, D.S. Pope, and A.M. Marcolini. Airfoil self-noise and prediction. Technical report, NASA RP-1218, July 1989.
94. J.S. Simonoff, Smooting Methods in Statistics, Springer - Verlag, 1996.
95. I.Cheng Yeh, Modeling of strength of high performance concrete using artificial neural networks, Cement and Concrete Research. **28**, pp. 1797-1808, 1998.
96. D. Harrison and D.L. Rubinfeld, Hedonic prices and the demand for clean ai, J. Environ. Economics & Management **5**, pp. 81-102, 1978.
97. J.S. Simonoff, Smooting Methods in Statistics, Springer - Verlag, 1996.
98. Mackowiak, P.A., Wasserman, S.S., Levine, M.M., 1992. A critical appraisal of 98.6 degrees f, the upper limit of the normal body temperature, and other legacies of Carl Reinhold August Wunderlich. J. Amer. Med. Assoc. 268, 1578–1580
99. R.D. King, S. Muggleton, R. Lewis, M.J.E. Sternberg, Proc. Nat. Acad. Sci. USA **89**, pp. 11322–11326, 1992.
100. M. Sikora, L. Wrobel, Application of rule induction algorithms for analysis of data collected by seismic hazard monitoring systems in coal mines, Archives of Mining Sciences **55**, pp. 91-114, 2010.
101. Tsoulos, I. G., Charilogis, V., Kyrou, G., Stavrou, V. N., & Tzallas, A. (2025). OPTIMUS: A Multidimensional Global Optimization Package. Journal of Open Source Software, 10(108), 7584.
102. Sanderson, C., & Curtin, R. (2016). Armadillo: a template-based C++ library for linear algebra. The Journal of Open Source Software, 1(2), 26-26.

103. Hogan, R. J. Adept C++ Software Library: User Guide.
104. Yuan, Y. X. (1991). A modified BFGS algorithm for unconstrained optimization. IMA Journal of Numerical Analysis, 11(3), 325-332.
105. C. Bishop, Neural Networks for Pattern Recognition, Oxford University Press, 1995.
106. G. Cybenko, Approximation by superpositions of a sigmoidal function, Mathematics of Control Signals and Systems **2**, pp. 303-314, 1989.
107. D. P. Kingma, J. L. Ba, ADAM: a method for stochastic optimization, in: Proceedings of the 3rd International Conference on Learning Representations (ICLR 2015), pp. 1–15, 2015.
108. Y. Xue, Y. Tong, F. Neri, An ensemble of differential evolution and Adam for training feed-forward neural networks. Information Sciences **608**, pp. 453-471, 2022.
109. K. O. Stanley, R. Miikkulainen, Evolving Neural Networks through Augmenting Topologies, Evolutionary Computation **10**, pp. 99-127, 2002.
110. S. Ding, L. Xu, C. Su et al, An optimizing method of RBF neural network based on genetic algorithm. Neural Comput & Applic 21, pp. 333–336, 2012.